AYUSH JHA

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Postdoctoral Scholar in Quantitative Finance & Financial Economics
1108 Memorial Circle, Department of Mathematics & Statistics, Texas Tech University
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Education

Ph.D., Economics, Texas Tech University, 2026
B.A. (Hons), Political Science & Economics, University of Delhi, 2022

Teaching and Research Fields

Fields: Asset Pricing, Financial Econometrics, Household Finance, Derivatives and Fixed-Income Valuation, Portfolio Choice and Risk Management, Financial Markets, Implications of Artificial Intelligence in Finance.

Scholarships and Awards

2026	Nominated for 2nd Abu Dhabi Investment Authority (ADIA) Financial Data Lab Best Paper Award
2024	Graduate Student Research Support Award
2023–now	Dr. Rashid B. Al-Hmoud Scholarship in Economics, Texas Tech University
2023–now	Presidential Doctoral Scholarship
2022	Graduate School Enhancement Scholarship, Texas Tech University

Media Representation/Invited Talks

- KCBD Television (NBC Lubbock): *Prediction markets are taking over sports gambling, here's how they work*

  On how platforms like Polymarket price sports outcomes, and why large traders move those prices, citing my joint work on belief formation and composition in prediction markets. [Link to paper]

- Office of Communications UK (Invited talk to EAG seminar team): Invited, with my co-authors, to present our study titled *"The Live-Nation and Ticketmaster Saga: A Robust Meger Analysis"* with Noah Liptack, Ali Jaffri, and Michael D. Noel. [Link to paper]

Job Market Paper

"Drift and Shape in Option-Implied Equity Pricing Kernels" with Papazyan and Rachev. [Link]

The maturity profile of an equity pricing kernel combines changes in its shape with changes induced by the equity premium. We derive an exact decomposition of the log kernel that separates these components when the physical return density is generated by a drift-invariant simulator. The decomposition yields computable bounds and identified sets for shape measures when the physical drift is uncertain, together with inference for the horizon profile. In S&P 500 index options, the premium-induced component accounts for most of the decline in a sign-based measure between one month and one year. By contrast, the detrended amplitude of the interior hump provides little evidence of economically large decay. The results illustrate how uncertainty about the physical drift affects the interpretation and statistical analysis of option-implied pricing kernels. We assess first-stage estimation error and study finite-sample performance in a Monte Carlo design calibrated to the empirical application with overlapping return horizons.

Research Experience and Other Employment

2026–now	Texas Tech University (TTU), Department of Mathematics & Statistics, Postdoctoral Scholar in Quantitative Finance
2023–2026	Texas Tech University (TTU), Department of Economics, Graduate Part-Time Instructor

Teaching Experience

Instructor

2026–2027	MATH 2345, Introduction to Statistics with Applications to Business (TTU)
2027	MATH 3356, Quantitative Theory of Interest (TTU)
2026	ECO 4316, Managerial Economics with Financial Data Analysis (TTU)
2025–2026	ECO 2301, Principles of Microeconomics (TTU)
2025	ECO 2305, Principles of Economics (TTU)

Teaching Assistant

2024	ECO 2305, Principles of Economics (TTU)
2024	ECO 2301, Principles of Microeconomics (TTU)
2023–2024	ECO 3312, Intermediate Microeconomics (TTU); taught the mathematics camp covering optimization and consumer theory by consent of instructor
2023	ECO 3302, Intermediate Macroeconomics (TTU)
2023	Economics Tutoring Center for undergraduates (TTU)

Intellectual Contributions

Publications

- Ayush Jha, Ali Jaffri, and Hassan A. Butt, "Household Interest in Artificial Intelligence for Financial Advice: Evidence from the United States." *Journal of Behavioral and Experimental Finance*, 2026, Vol. 51, 101212.
- Abigail Mensah, Ayush Jha, Hongwei Mei, Rui Wang, Svetlozar Rachev, and Frank J. Fabozzi, "Option Pricing under Stochastic Volatility and Jumps: A PIDE Framework with Empirical Evidence." *Frontiers of Mathematical Finance*, 2026, Vol. 9, pp. 41–66.
- Ayush Jha, "Reinforcement Learning for Strategic Asset Allocation: Why the Objective Function Dominates the Agent Architecture." *Journal of Financial Data Science*, 2026.
- Ayush Jha, Ali Jaffri, Svetlozar T. Rachev, and Frank J. Fabozzi, "Option-Implied Probabilities and Bond Valuation." *The Journal of Fixed Income*, 2026, Vol. 35(3), pp. 6–17.
- Ayush Jha, Ali Jaffri, Svetlozar T. Rachev, and Frank J. Fabozzi, "Equity-Imposed Tilts in Affine Term Structure Models: Evidence from Option-Implied Asymmetries." *The Journal of Fixed Income*, 2026, Vol. 35(4), pp. 7–24.
- Ayush Jha, Ali Jaffri, and Hassan A. Butt, "Demystifying FinBERT: How Transformer Models Turn Financial Text into Market Insights." *The Journal of Portfolio Management*, 2025, Vol. 52(2), p. 212.
- Ayush Jha, Abootaleb Shirvani, Svetlozar T. Rachev, and Frank J. Fabozzi, "Beyond the Traditional VIX: A Novel Approach to Identifying Uncertainty Shocks in Financial Markets." *Journal of Risk and Financial Management*, 2025, Vol. 18(1), 11.
- Ali Jaffri, Abootaleb Shirvani, Ayush Jha, Svetlozar T. Rachev, and Frank J. Fabozzi, "Optimizing Portfolios with Pakistan-Exposed Exchange-Traded Funds: Risk and Performance Insight." *Journal of Risk and Financial Management*, 2025, Vol. 18(3), 158.

Working Papers

- Ayush Jha, "Distributional Granger Causality: Identification, Sequential Inference, and Adaptive Testing." *(R&R – Econometric Reviews)*
- Ali Jaffri, Ayush Jha, and Hassan A. Butt, "Social Media as an Investment Information Channel: Evidence from U.S. Households." *(R&R – Global Finance Journal)*
- Emma Burke, Ayush Jha, Svetlozar T. Rachev, and Frank J. Fabozzi, "The Common Advice: Evidence on AI Redundancy in Stock Prices."
- Ayush Jha, "Household Financial Planning in General Equilibrium: Theory and Micro Evidence."
- Ayush Jha, Ali Jaffri, and Hassan A. Butt, "Survivorship Bias in the Cross-Section of Winner Portfolios."
- Ayush Jha, Abootaleb Shirvani, Ali Jaffri, Svetlozar T. Rachev, and Frank J. Fabozzi, "Recursive Portfolio Learning and Endogenous Riskless Rate."
- Ayush Jha, "Agentic AI and Real-Time Market Information Premium."

- Ayush Jha, Abootaleb Shirvani, Ali Jaffri, Svetlozar T. Rachev, and Frank J. Fabozzi, "State-Dependent Portfolio Choice under Policy-Regime Switching: Evidence from Environmental, Social, and Governance Momentum Strategies."

Books and Monographs

- Jha, A., Jaffri, A., Lindquist, W.B., Rachev, S., Boakye, R., Hettiachchi-Halpe-Kankanamalage, D.C.W., & Mensah, A. (2027). *Microstructure, Statistical Fluctuations, and Technical Signals in Rational Asset Pricing*. 1st ed., Chapman and Hall/CRC.

Conferences and Competitions

Southern Economic Association Annual Meetings, Houston (Fall 2026), Midwest Macro Meetings, Lubbock (Fall 2026), 19th International Behavioral Finance Conference, Booth School of Business, University of Chicago (Fall 2025), Texas Economic Theory Camp, SMU (Fall 2025), ISA-South, Lubbock (Fall 2025), Eastern Economic Association Annual Meetings, New York City (Spring 2025), Graduate Student Research Poster Competition, Lubbock (Spring 2024), Midwest Macroeconomics Conference, Lubbock (Fall 2023)

Software

Bloomberg Terminal; Wharton Research Data Services (WRDS)—CRSP, OptionMetrics, Compustat; Python; MATLAB; High Performance Computing Centre.

Professional Service

Referee/Reviewer: *Global Finance Journal*, *Finance Research Letters*, *Financial Innovation*, *Frontiers of Artificial Intelligence*.

References

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